

Krishna
Online ML, DL, NLP
course

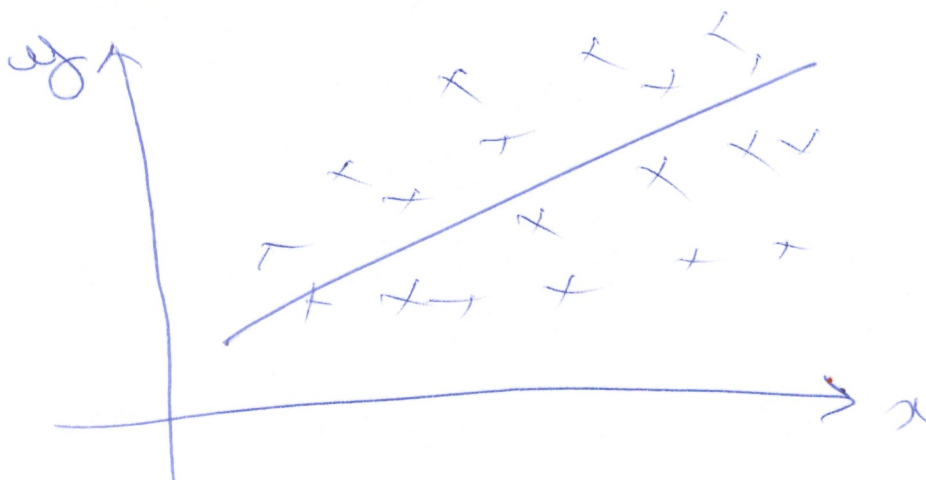
①

Sec - 26

Understanding complete
Linear Regression Intuition

lec - 13

Polynomial Regression
Intuition



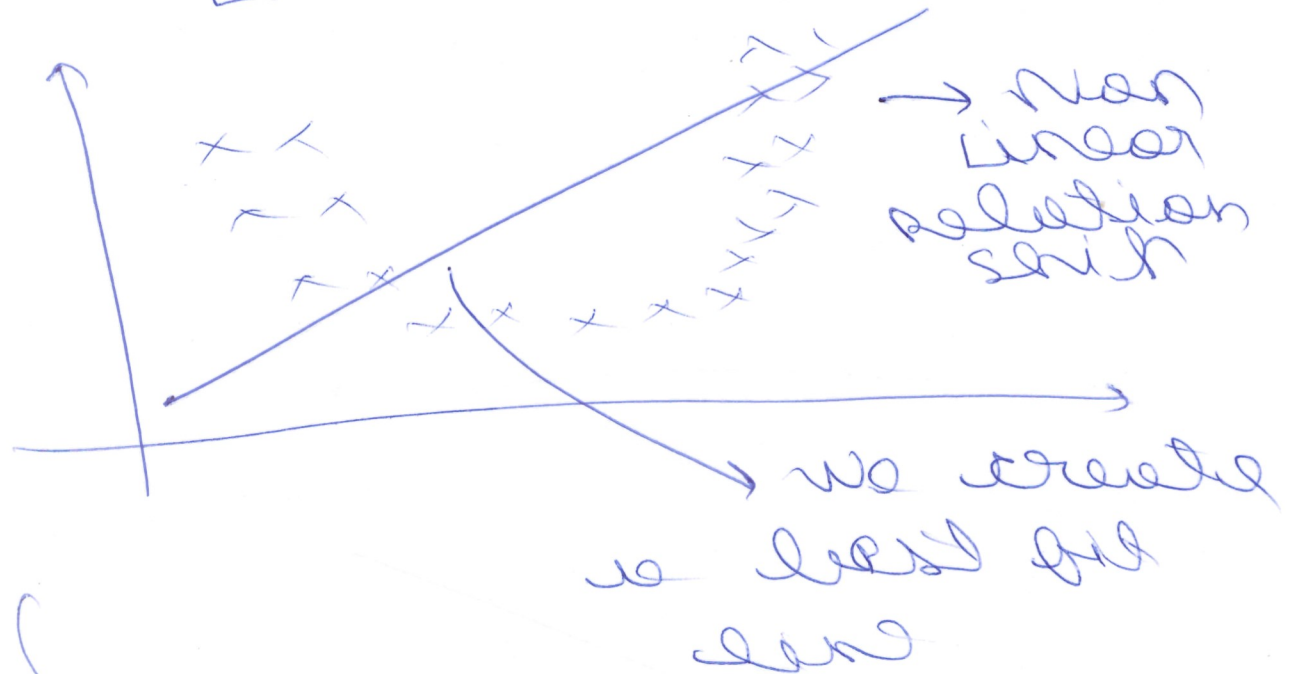
$$h_0(x) = \beta_0 + \beta_1 x$$

↳ simple linear regression

$$h_0(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

↳ multiple linear regression

In polynomial regression, the graph, there's no linear relation



→ there the errors will be quite high.

Hence there's going to be a lot of error.

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Now our way must
be plot a ~~best~~ fit
line :-

such that the
error is minimized.

→ this will be done as:



we will try polynomial
regression.

⇒ First let's see polynomial
of degree

degree = 0.

⇒ Simple polynomial regression
with degree = 0

$$h_0(x) = \beta_0 + \beta_1 x$$

degree = 0

polynomial degree =

$$h_0(x) = \beta_0 x^0 + \beta_1 x^1$$

$$\therefore h_0(x) = \beta_0 x^0$$

polynomial degree = 0 $\Rightarrow$ This will be a constant value.

$\Rightarrow$ Now suppose, the polynomial degree = 1

$$h_0(x) = \beta_0 x^0 + \beta_1 x^1$$

$\swarrow$
This is basically simple Linear regression.

(*) Suppose if polynomial degree = 2

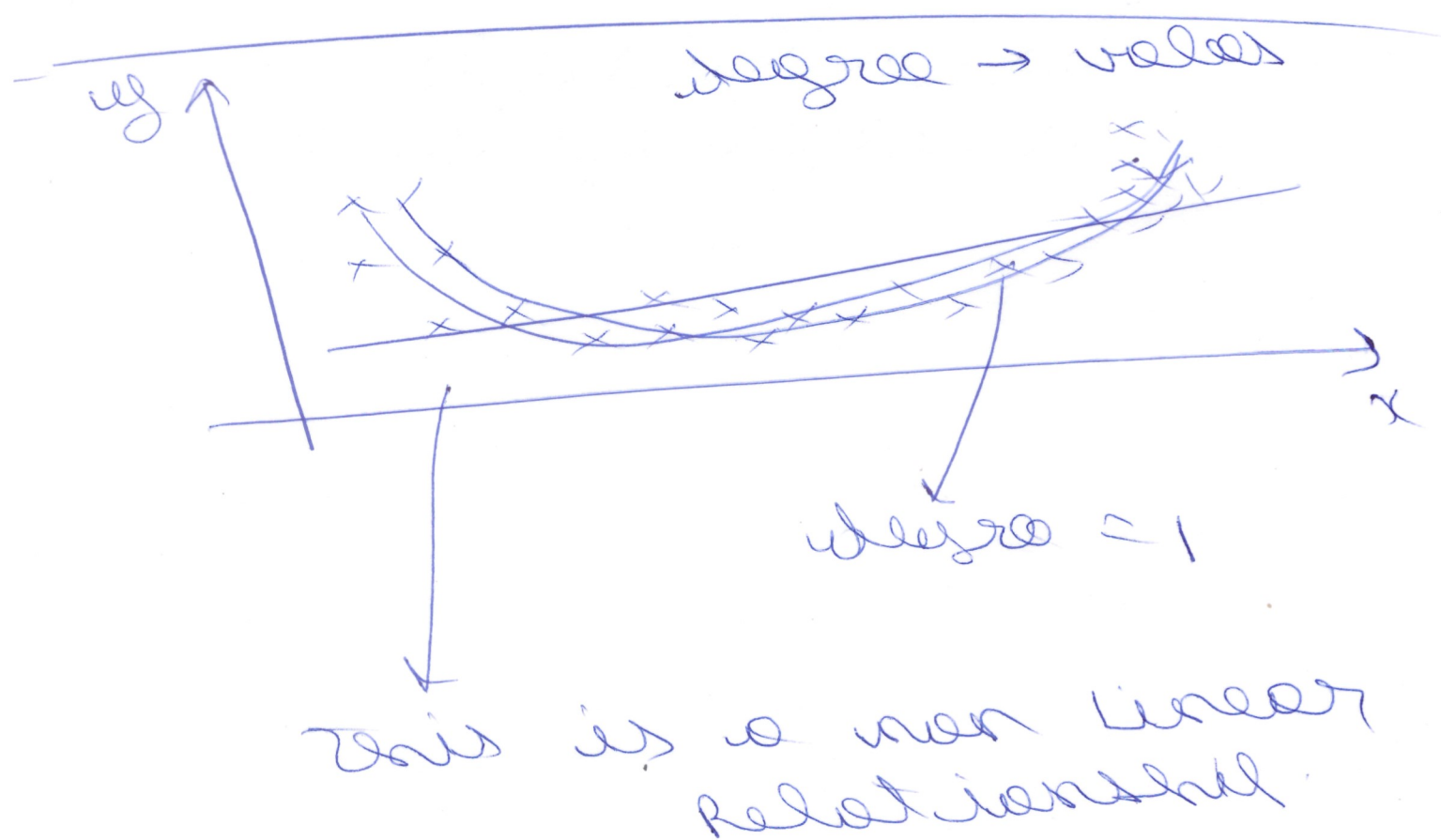
$$h_0(x) = \beta_0 x^0 + \beta_1 x^1 + \beta_2 x^2$$

$\swarrow$
Here this is degree = 2 polynomial regression

Linear polynomial
~~the~~ degree = 2

(5)

$$\ln(x) = \beta_0 x^0 + \beta_1 x^1 + \beta_2 x^2 + \dots + \beta_n x^{(n)} \text{ Ans}$$



(*) Suppose, if we have more than 1 independent features

Degree = 1

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$$h_0(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Degree = 2

$$h_0(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2$$

→ For multiple independent scalars